Acoustic Occupancy Modelling of Bats at Wind Farms: Integrating Effort, Environmental Covariates, and Spatial Cross-Validation

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Abstract—Wind energy and bat conservation intersect where turbines are sited in landscapes used by foraging and migratory bats. Passive acoustic monitoring (PAM) can characterise bat occurrence at fine spatial and temporal scales, but imperfect detection, species misidentification, effort, and spatial autocorrelation may introduce inference. Occupancy models explicitly separate latent occurrence from detection, making them a principled choice for presence-based inference from acoustics. This literature review synthesises key contributions on occupancy modelling, misclassification, spatial extensions, validation strategies, and bat-wind-farm impacts using a focused set of sources.

1 Introduction

Wind farms can elevate mortality for migratory and tree-roosting bats, with fatalities peaking seasonally and under specific wind regimes. Reviews of North American wind facilities show that fatalities are dominated by migratory lasiurine species and tend to peak in late summer and autumn, often under low wind speeds and particular weather conditions [1, 2, 3, 4]. These patterns raise concern that wind energy development may increase population-level risks for some bat species [4].

Passive acoustic monitoring (PAM) offers a scalable, non-invasive way to monitor bat activity and presence around turbines. However, acoustic detections reflect both ecological state and observation process: instruments differ in sensitivity, ambient conditions vary, and automated or manual classification can introduce misidentifications [5]. Occurrence must instead be estimated while accounting for imperfect detection and potential false positives.

Occupancy models provide a hierarchical framework that separates ecological occurrence (probability a site is used) from detection (probability of recording a species given it is present) [6, 7]. Extensions accommodate dynamic processes such as colonization and local extinction [8], misidentification in detection data [9, 10], and spatial autocorrelation across sites [11]. Coupled with modern cross-validation strategies that respect spatial and temporal structure [12, 13], this framework is well suited to acoustic monitoring around wind farms.

This review uses a focused set of key sources to summarise occupancy modelling concepts relevant to bat acoustics, discuss misclassification-aware and spatial occupancy models, relate landscape and environmental drivers of bat occurrence and fatalities to turbine contexts, and outline validation approaches that reduce optimistic bias in performance estimates for spatially structured data.

2 Foundations of Occupancy Modelling

2.1 Single-season occupancy

Single-season occupancy models estimate site-level occurrence (ψ) and detection probability (p) from replicate detection/non-detection surveys at each site [6, 7]. The ecological state z_i at site i is modelled as a Bernoulli variable with parameter ψ_i, potentially influenced by covariates such as habitat or distance to turbines. Conditional on z_i, detection histories y_{ij} for visit j are Bernoulli with parameter p_{ij}.

MacKenzie et al. show that failing to account for imperfect detection leads to biased estimates of occupancy, particularly when detection probabilities are low or vary across sites and visits [6]. Their simulations and case studies demonstrate that even modest detection probabilities can result in substantial underestimation of ψ if raw detection frequencies are interpreted as occupancy. This is directly relevant for acoustic surveys in which detection is influenced by weather, equipment, and species-specific call properties [5]. Figure 1 summarises these relationships in a hierarchical framework.

2.2 Dynamic occupancy

Dynamic (multi-season) occupancy models extend the framework to multiple primary periods, allowing explicit estimation of colonization (γ) and local extinction (ε) between periods [7, 8]. The latent state evolves as a Markov process, with occupancy in each period determined by previous state and transition probabilities. MacKenzie et al. applied this to long-term monitoring data for spotted owls and salamanders, showing that dynamic occupancy can be used to estimate vital rates of metapopulations when detection is imperfect [8]. For bat acoustics at wind farms, such dynamic models could describe how occupancy patterns shift across seasons (e.g., migration, breeding) and across years in response to turbine operation.

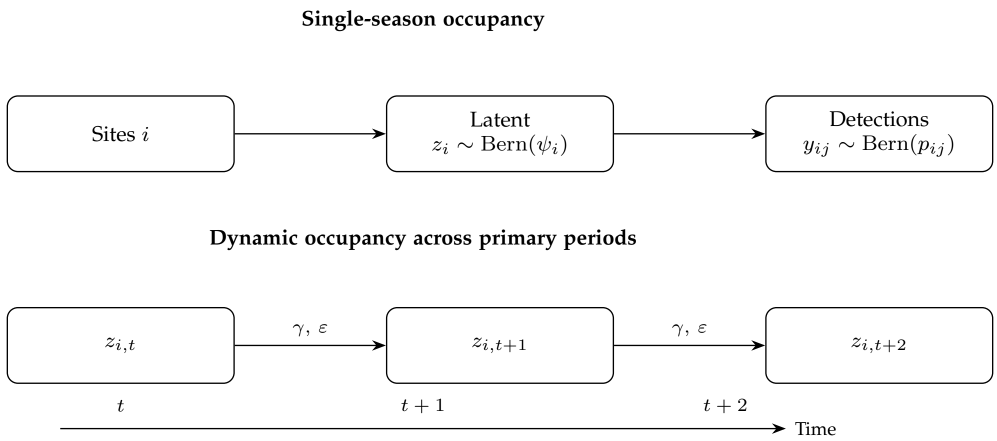

Fig. 1. Conceptual diagram of single and dynamic occupancy models. Top: single-season structure where latent occupancy z_i with probability ψ_i generates detections y_{ij} with detection probability p_{ij}. Bottom: dynamic occupancy where the latent state evolves across primary periods via colonization γ and local extinction ε.

3 Misclassification and False-Positive Occupancy

Conventional occupancy models assume that when a species is detected it is correctly identified, so false positives do not occur [6, 7]. However, automated acoustic classifiers and even expert interpreters can misidentify calls, particularly among species with similar echolocation characteristics. Treating all detections as error-free can bias occupancy estimates and covariate effects.

Miller et al. propose models that simultaneously account for non-detection (false negatives) and misidentification (false positives) [9]. Their approach extends the detection process to distinguish three outcomes: true detection, false-positive detection, and non-detection, with parameters informed by a subset of observations for which misidentification is assumed absent or negligible. Simulations and frog call data illustrate that ignoring misidentification can lead to substantial bias in occupancy, colonization, and extinction estimates, whereas misclassification-aware models recover unbiased parameters at the cost of increased uncertainty [9].

Chambert et al. generalise this framework to a range of study designs that include ambiguous detections and varying levels of known-truth information [10]. They compare designs in which validation data are collected at the site level versus the observation level, and provide guidance on how to choose among them. Their work underscores that modelling strategy must be aligned with the validation design, and that different designs yield different trade-offs between bias and variance. For turbine-scale acoustic occupancy studies, these results highlight the value of allocating effort to manual verification of a subset of automated detections so that misclassification can be explicitly modelled.

4 Spatial Occupancy Models

Ecological processes and observational conditions are spatially structured, leading to spatial autocorrelation in occupancy and detection. Conventional occupancy models assume that sites are conditionally independent given covariates [6, 7]. When this assumption is violated, standard errors can be underestimated and spatial patterns mischaracterized.

Johnson et al. present spatial occupancy models designed for large datasets, incorporating spatial random effects into the occurrence process [11]. Their approach uses Gaussian random fields or related structures to capture residual spatial dependence beyond what covariates explain. They demonstrate that spatial occupancy models can handle very large datasets and provide improved inference about spatial patterns of species occurrence compared with non-spatial models. Such models are appealing for wind farms with dense detectors or numerous turbines, where residual spatial structure is likely.

Royle and Dorazio provide a broader hierarchical modelling perspective, treating spatial occupancy models as one instance of hierarchical models that account for multiple sources of variability and correlation [7]. Their framework supports Bayesian inference and model comparison, which are useful when integrating spatial random effects, misclassification, and dynamic processes in a single model.

5 Bat Occupancy, Habitat, and Wind-Energy Impacts

5.1 Landscape structure and habitat associations

Forest structure and fragmentation influence bat occupancy and activity, with implications for turbine siting in forested landscapes. Yates examined bat occupancy in Missouri Ozark forests and found that forest structure and fragmentation had species-specific effects on site occupancy [14]. Some species were more likely to occupy less fragmented, structurally complex forests, highlighting the importance of retaining key habitat elements.

Ekman and de Jong used bat detectors to examine the distribution and resource use of four bat species in patchy and continuous environments [15]. They found that a crop-field matrix had a stronger negative influence on bat occurrence than water, and that certain species were negatively affected by isolation of forest patches. These results suggest that landscape configuration, not just habitat amount, shapes bat occurrence, and that some species may

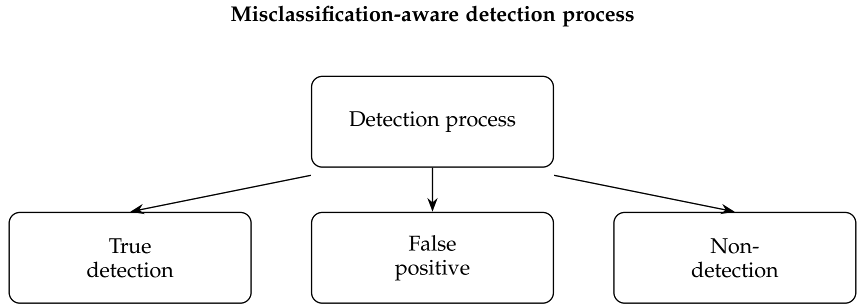

Fig. 2. Schematic of a misclassification-aware detection process. The observation model distinguishes among true detections, false-positive detections, and non-detections, allowing occupancy models to account for both false negatives and false positives in acoustic data.

TABLE 1
Summary of key occupancy and validation model types relevant to turbine-scale bat acoustics. FP = false positives; dyn. = dynamic processes; spat. = spatial dependence.

Model / reference	Main idea	Handles FP?	Handles dyn.?	Handles spat.?	Typical data requirements	Relevance to bat acoustics at wind farms
MacKenzie et al. 2002 [6]	Single-season occupancy with imperfect detection	No	No	No	Repeated detections/non-detections at sites within a season	Baseline framework for turbine-level or detector-level occurrence with replicate nights.
MacKenzie et al. 2003 [8]	Dynamic occupancy with colonization and extinction	No	Yes	No	Repeated surveys across multiple primary periods (seasons/years)	Captures seasonal and interannual changes in bat use of turbine areas.
Royle & Dorazio 2008 [7]	Hierarchical modelling perspective on occupancy	No	Yes (general)	Possible (general)	Flexible hierarchical data structures, often Bayesian	Provides a unifying framework to combine detection, dynamics, misclassification, and spatial effects.
Miller et al. 2011 [9]	False-positive occupancy models	Yes	Possible	No	Subset of validated detections and ambiguous automated classifications	Allows explicit modelling of misidentified bat calls in acoustic datasets.
Chambert et al. 2015 [10]	General misclassification-aware designs	Yes	Possible	No	Study designs with site-level or observation-level validation data	Guides how to allocate manual verification effort for acoustic detections.
Johnson et al. 2013 [11]	Spatial occupancy with random fields for large datasets	No (by default)	Possible	Yes	Large numbers of spatially referenced sites	Accounts for residual spatial structure in turbine arrays or detector networks.
Roberts et al. 2017 [12]	Cross-validation for structured ecological data	N/A	N/A	N/A	Any model with temporal/spatial/phylogenetic structure	Recommends block cross-validation to avoid optimistic performance estimates.
Valavi et al. 2019 [13]	`blockCV` for spatial/environmental blocking	N/A	N/A	N/A	Point data with coordinates and covariates	Provides practical tools for implementing spatial or environmental blocks for validation of occupancy models.

be particularly vulnerable to increased patchiness and open-field matrices.

Hayes provided a detailed discussion of assumptions and practical considerations in echolocation-monitoring studies, emphasizing that acoustic detections are influenced by habitat structure, clutter, and survey design [5]. Taken together, these studies support the use of habitat and fragmentation covariates in occupancy models for bat acoustics, and suggest that turbine placement near certain habitat edges or corridors may interact with occupancy patterns.

5.2 Wind-energy impacts and environmental drivers

Several studies synthesize patterns of bat fatalities at wind-energy facilities. Arnett et al. review post-construction fatality studies across North America and report that bat fatalities are dominated by migratory, foliage- and tree-roosting species, with fatalities peaking in late summer and fall [1]. Most fatalities occur on nights with low wind speeds and in association with passing storm fronts. They caution that fatality estimates are sensitive to survey design and biases in carcass detection, but nonetheless identify consistent patterns across sites.

Baerwald and Barclay used acoustic monitoring and carcass searches at an Alberta wind facility to examine temporal and spatial variation in bat activity and fatalities [2]. They found that overall activity of migratory bats increased at low wind speeds and warm temperatures, and that fatalities were influenced by weather variables such as barometric pressure and moon illumination. Species differed in their responses to environmental conditions, and turbine location within the facility affected fatality risk. These results illustrate the complex interplay between weather, turbine operation, and bat behaviour.

Frick et al. used expert elicitation and population projection models to assess whether turbine-related fatalities could threaten the population viability of hoary bats [4].

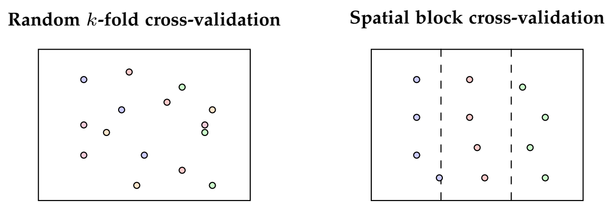

Fig. 3. Illustration of cross-validation schemes for turbine-scale acoustic occupancy models. Left: random k-fold cross-validation assigns turbines to folds without regard to spatial structure, risking overly optimistic performance estimates. Right: spatial block cross-validation groups turbines into contiguous spatial blocks, better mimicking prediction to new locations.

Their models suggest that current levels of mortality could lead to substantial population declines and elevated extinction risk. Although focused on population-level impacts rather than occupancy per se, these findings underline the importance of accurately characterizing occurrence and activity patterns around turbines as a step toward assessing risk.

Cryan and Barclay synthesise hypotheses for the proximate and ultimate causes of bat fatalities at wind turbines, distinguishing random, coincidental, and attraction-driven collisions [3]. They argue that some hypotheses imply specific behavioural or spatial patterns (e.g., aggregation during migration, attraction to turbines as potential roosts or feeding sites), which could be investigated with occupancy models that include relevant covariates.

6 Cross-Validation and Model Evaluation

Ecological data often exhibit spatial and temporal structure, and standard random cross-validation can severely underestimate predictive error [12]. Roberts et al. review cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure, showing via simulations and case studies that block cross-validation is generally more appropriate than random folds when predicting to new locations or times [12]. They emphasise that dependence structures may remain in model residuals even when correlation is modelled, and that blocking must be carefully chosen to reflect the intended prediction task.

Valavi et al. introduce the `blockCV` package, which implements spatial and environmental blocking for k-fold cross-validation in species distribution modelling [13]. The package provides tools to measure spatial autocorrelation in covariates and to generate folds that are separated in space or environmental space, helping to avoid overly optimistic performance estimates. Although `blockCV` was designed for species distribution models, the same principles apply to occupancy models with spatially structured data.

For turbine-scale acoustic occupancy studies, these works suggest that validation should use spatial or spatio-temporal blocks that separate turbines or clusters of detectors, rather than random assignment of site-visits to folds. This is particularly important when models are used to predict to new turbines or future periods, where random cross-validation would not reflect the real extrapolation task. Figure 3 provides a schematic comparison of random versus blocked validation strategies. Combined with the methodological summary in Table 1, this clarifies how choice of validation scheme interacts with model structure and intended use.

7 Application to Acoustic Occupancy at Wind Farms

Drawing on the reviewed literature, a practical workflow for acoustic occupancy modelling around wind farms can be outlined:

1) **Survey design:** Define sites as turbines or spatial units (e.g., buffers or grid cells) that satisfy approximate closure assumptions over the primary period [6, 8]. Plan multiple nightly visits within each primary period to support estimation of detection probability.
2) **Covariate selection:** Include habitat and fragmentation metrics inspired by Yates and Ekman and de Jong to capture landscape influences on occupancy [14, 15]. Add turbine-related and weather covariates shown to affect bat activity and fatalities, such as wind speed, temperature, barometric pressure change, and moon illumination [1, 2, 4]. Table 2 summarises these empirical drivers and how they can inform covariate choice.
3) **Acoustic data processing:** Apply quality control and conservative classification thresholds, recognising potential misidentification and the detection biases discussed by Hayes [5]. Where possible, generate a subset of manually validated detections to support misclassification-aware models.
4) **Modelling:** Fit single- or dynamic occupancy models that include occurrence and detection covariates, building on the frameworks of MacKenzie et al. and Royle and Dorazio [6, 7, 8]. Where misidentification is a concern, consider false-positive occupancy models following Miller et al. and Chambert et al. [9, 10]. For dense or large-scale arrays, incorporate spatial random effects as in Johnson et al. to account for residual spatial structure [11]. The main options and their trade-offs are summarised in Table 1.
5) **Validation:** Use spatial or spatio-temporal block cross-validation, potentially implemented with tools such as `blockCV`, to obtain more realistic estimates of predictive performance [12, 13]. Evaluate both discrimination (e.g., AUC) and calibration, and examine ecological plausibility of predictions, following the conceptual guidance in Figure 3.

Such a pipeline integrates methodological advances in occupancy modelling with empirical knowledge about bat behaviour and turbine impacts, and provides a basis for producing occurrence maps and turbine-level risk indices.

8 Limitations and Research Needs

Several limitations emerge from this review. First, most occupancy modelling work has focused on non-bat taxa and on survey methods other than acoustics; adapting these models to bat acoustics requires attention to unique detection processes and misclassification patterns [5, 9]. Second,

TABLE 2
Selected empirical studies on bat occupancy, habitat associations, and wind-energy impacts, and their implications for turbine-scale acoustic occupancy modelling.

Study	Focal taxa / region	Key predictors examined	Main effects on occupancy/activity/fatalities	Implications for turbine-scale occupancy models
Yates 2010 [14]	Forest bats, Missouri Ozarks	Forest structure, fragmentation metrics	Species-specific responses; some species favour less fragmented, structurally complex forests	Include fragmentation and forest-structure covariates when modelling occupancy near forested turbines.
Ekman & de Jong 1996 [15]	Four bat species in patchy vs continuous environments	Patch isolation, matrix type (crop vs water)	Crop-field matrix more negative than water; isolation reduces occurrence for some species	Represent landscape configuration and matrix type around turbines as covariates.
Hayes 2000 [5]	Echolocating bats, multiple systems	Habitat clutter, detector placement, survey design	Detection strongly affected by habitat structure and methods	Model detection as a function of habitat and survey design; interpret non-detections cautiously.
Arnett et al. 2008 [1]	Bats at North American wind facilities	Season, wind speed, weather conditions	Fatalities dominated by migratory tree bats; peaks in late summer/fall and low wind conditions	Align primary periods and covariates with seasonal migration and low-wind regimes.
Baerwald & Barclay 2011 [2]	Alberta wind facility	Wind speed, temperature, barometric pressure, moon illumination, turbine position	Migratory activity higher at low wind and warm temperatures; fatalities vary with weather and turbine location	Include fine-scale weather and turbine-location covariates in occurrence and detection submodels.
Frick et al. 2017 [4]	Hoary bats, North America	Fatality rates at turbines, demographic parameters	Projected substantial population declines under current fatality rates	Use occupancy maps to identify high-use areas where mitigation could reduce population-level risk.
Cryan & Barclay 2009 [3]	Bats at wind turbines	Hypothesised causes of collisions (random vs attraction)	Evidence consistent with attraction for some species/contexts	Test attraction/avoidance hypotheses by including turbine proximity and operation in occupancy models.

spatial occupancy models remain relatively computationally demanding and may require careful approximation or dimension reduction for large wind farms [11]. Third, while cross-validation strategies for spatial data are now better understood, their application in occupancy contexts remains limited, and there is little consensus on standard practice for turbine-scale studies [12, 13].

From a conservation perspective, the evidence that turbine fatalities may meaningfully affect population trajectories for migratory bats [4] underscores the need for occupancy studies that explicitly link occurrence patterns to collision risk and mitigation strategies. Hypotheses about attraction or avoidance of turbines [3] could be tested with models that incorporate turbine proximity and operation as covariates, using acoustic occupancy as a proxy for collision risk.

9 Conclusion

Occupancy models offer a flexible and principled framework for interpreting acoustic detections of bats around wind turbines, addressing imperfect detection and, when extended, misidentification and spatial dependence. Foundational work on single and dynamic occupancy [6, 7, 8], misclassification-aware models [9, 10], and spatial occupancy [11] provides the methodological tools needed for turbine-scale analyses. Empirical studies of forest structure, fragmentation, and landscape matrices highlight the importance of habitat and configuration for bat occurrence [5, 14, 15], while syntheses of turbine fatalities and population viability emphasise the urgency of understanding bat–wind-farm interactions [1, 2, 3, 4].

Finally, modern cross-validation strategies tailored to structured ecological data [12, 13] are essential to ensure that occupancy models used for risk assessment and mitigation are evaluated honestly. Future work should integrate these components in applied studies, focusing on misclassification-aware, spatially explicit, and dynamically validated occupancy analyses that directly inform wind-energy planning and conservation decisions.

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